Online Resource 1 for "When Does Relational Structure Help Healthcare Fraud Detection? A Controlled Benchmark of Heterogeneous Graph Neural Networks and Graph-Augmented Tabular Models", *Health Information Science and Systems*

H M Akshay[1*], Gaurav S B[1], Eshita Narayana[1], Harshita Srivastava[1], Malini M Patil[1]

[1]Department of Computer Science and Engineering, RV Institute of Technology and Management, Bengaluru, 560083, Karnataka, India.

*Corresponding author(s). E-mail(s): rvit23bcs106.rvitm@rvei.edu.in;
Contributing authors: rvit23bcs258.rvitm@rvei.edu.in;
rvit23bcs031.rvitm@rvei.edu.in; rvit23bcs323.rvitm@rvei.edu.in;
malinimp.rvitm@rvei.edu.in;

Abstract

This Online Resource accompanies the article named in the title. It contains material summarised but not shown in full there: the selected hyperparameters, the two-stage routing evaluation, the ring-volume dose-response, the confirmatory test of the hybrid result, the cost of training and scoring, single-factor ablations of HGT, which inputs the trained models use, figures of the mechanism ablations and of new versus recurring rings, repeated cross-validation and per-rotation tuning on the public data, and a re-analysis of a small fixed benchmark. Every number is generated from the run records in the article's code repository.

S1 Selected hyperparameters

Table S1 lists the configurations selected with the equal tuning budget.

Table S1 Selected hyperparameters (equal budget per method; hybrids inherit their base configuration; methods sharing a search space were offered the same candidate configurations). Synthetic: tuned on development worlds; public data: tuned on a separate split. The configurations re-tuned for the low-volume stress test are in the repository (`results/tuning_lowvolume`).

Method	Synthetic benchmark	Public data
LR	C=100	C=1
RF	max depth=16; max features=sqrt; min samples leaf=3; n estimators=300	max depth=16; max features=0.3; min samples leaf=10; n estimators=300
GB (C)	l2=0; learning rate=0.03; max iter=200; max leaf nodes=15; min samples leaf=10	–
GB (C+A / P)	l2=1; learning rate=0.1; max iter=200; max leaf nodes=63; min samples leaf=50	l2=0; learning rate=0.03; max iter=200; max leaf nodes=15; min samples leaf=50
GB (C+A+G / P+G)	l2=1; learning rate=0.1; max iter=400; max leaf nodes=15; min samples leaf=20	l2=0; learning rate=0.03; max iter=200; max leaf nodes=15; min samples leaf=50
GB (P+G+L)	–	l2=0; learning rate=0.03; max iter=200; max leaf nodes=15; min samples leaf=50
MLP	dropout=0.3; hidden=128; layers=2; lr=0.001; weight decay=1e-05	dropout=0.3; hidden=128; layers=2; lr=0.001; weight decay=1e-05
SAGE	aggr=sum; dropout=0.1; hidden=32; layers=3; lr=0.003; weight decay=0.001	aggr=mean; dropout=0.3; hidden=64; layers=3; lr=0.003; weight decay=0.0001
RGCN	aggr=sum; dropout=0.3; hidden=64; layers=2; lr=0.003; weight decay=0.001	aggr=sum; dropout=0.3; hidden=64; layers=2; lr=0.01; weight decay=0.001
HAN	dropout=0.3; heads=4; hidden=32; layers=3; lr=0.01; weight decay=1e-05	dropout=0.1; heads=2; hidden=64; layers=2; lr=0.003; weight decay=0.0001
Simple-HGN	dropout=0.3; heads=4; hidden=64; layers=3; lr=0.003; weight decay=0.0001	dropout=0.1; heads=4; hidden=64; layers=2; lr=0.01; weight decay=0.001
HGT	dropout=0.3; heads=4; hidden=32; layers=3; lr=0.01; weight decay=1e-05	dropout=0.1; heads=4; hidden=64; layers=2; lr=0.01; weight decay=0.001

S2 Two-stage routing

S2.1 Method

A common deployment proposal is to score every claim with a cheap tabular model and pass only high-risk claims to a GNN. We evaluate this with GB-CAG as the first stage and HGT or SAGE as the second. A claim is escalated if its GB-CAG score is at least the $(1 - e)$ quantile of the GB-CAG scores in the validation period. The escalation fraction $e \in \{0.05, 0.1, 0.2, 0.3, 0.5, 1\}$ is chosen by validation AP, so test claims never influence it. Escalated claims rank above all other claims and are ordered either by the GNN score alone (*re-rank*: GB-CAG acts as a filter) or by the mean of the two models' validation percentiles (*fused*); non-escalated claims keep their

GB-CAG order. With $e = 1$, re-ranking reduces to the GNN alone and fusion to a rank-average ensemble of the two models, so both single models and the ensemble are among the candidates. A claim's routed score depends only on its own two scores and on validation statistics. The stages are the models of the main experiment, retrained with the same seeds; their test predictions reproduce the stored ones exactly for the GNNs and to within 3×10^{-8} for GB-CAG (float32 storage).

S2.2 Results

Table S2 and Fig. S1 evaluate two-stage routing (Section S2.1) on the same test claims as the single models. In the design usually proposed, GB-CAG filters and the GNN decides (re-rank). Validation mostly selected small escalation fractions (5% in 30 of 54 worlds for HGT), and the routed system matched GB-CAG (GB-CAG→HGT ΔAP $= -0.001$, 95% CI $[-0.006, 0.003]$; GB-CAG→SAGE $+0.001$). Escalating more claims to HGT lowered both ring AP (from 0.980 at 5% to 0.968 when every claim is escalated) and, more steeply, isolated AP (from 0.626 to 0.508; Fig. S1b–d), because HGT ranks both less well than GB-CAG. Ordering escalated claims by the mean of the two models' percentiles did better: GB-CAG→HGT gained $+0.003$ (Holm $p = 0.60$) and GB-CAG→SAGE $+0.006$ $[0.003, 0.010]$ (Holm $p = 0.042$, positive for all six seeds), on ring ($+0.006$) and isolated fraud ($+0.023$) alike. Validation, however, chose to fuse *every* claim in 22 (HGT) and 30 (SAGE) of the 54 worlds, and on average 55% and 70% of test claims were escalated: the gain comes from combining two different rankings, not from routing a subset of claims. The feature-level hybrids, which give the GNN the engineered features, had slightly higher ring AP ($+0.008$, not significant with the seed as unit) but lost 0.08–0.11 isolated AP and so matched GB-CAG overall. At the ring level all methods reached the ceiling: in all 54 worlds each test ring had at least one claim among the top 5% of claims, and precision in that budget was 0.989–0.998 for every method.

Table S2 Two-stage routing versus single models and feature-level hybrids, averaged over the 54 worlds (same test claims for every row). Routing escalates claims whose GB-CAG score exceeds a threshold set on validation data; the escalation fraction is chosen by validation AP. P@5%: precision among the top 5% of test claims. Esc.: mean share of test claims escalated. ΔAP: paired difference in AP from GB-CAG ($\pm$ half-width of the 95% t-interval over the six generator seeds), Holm-adjusted p over the rows.

Method	AP	Ring AP	Isol. AP	P@5%	Esc.	ΔAP (p)
GB-CAG	0.947	0.979	0.626	0.996	–	–
HGT	0.929	0.968	0.507	0.989	–	-0.018 ± 0.013 (0.104)
SAGE	0.940	0.979	0.541	0.995	–	-0.007 ± 0.009 (0.597)
HGT+AG	0.945	0.986	0.514	0.996	–	-0.002 ± 0.006 (1.000)
SAGE+AG	0.947	0.987	0.548	0.996	–	0.000 ± 0.005 (1.000)
GB-CAG→HGT, re-rank	0.945	0.979	0.606	0.995	0.16	-0.001 ± 0.005 (1.000)
GB-CAG→HGT, fused	0.950	0.981	0.640	0.997	0.55	$+0.003 \pm 0.004$ (0.597)
GB-CAG→SAGE, re-rank	0.948	0.983	0.595	0.997	0.24	$+0.001 \pm 0.004$ (1.000)
GB-CAG→SAGE, fused	0.953	0.985	0.648	0.998	0.70	$+0.006 \pm 0.004$ (0.042)

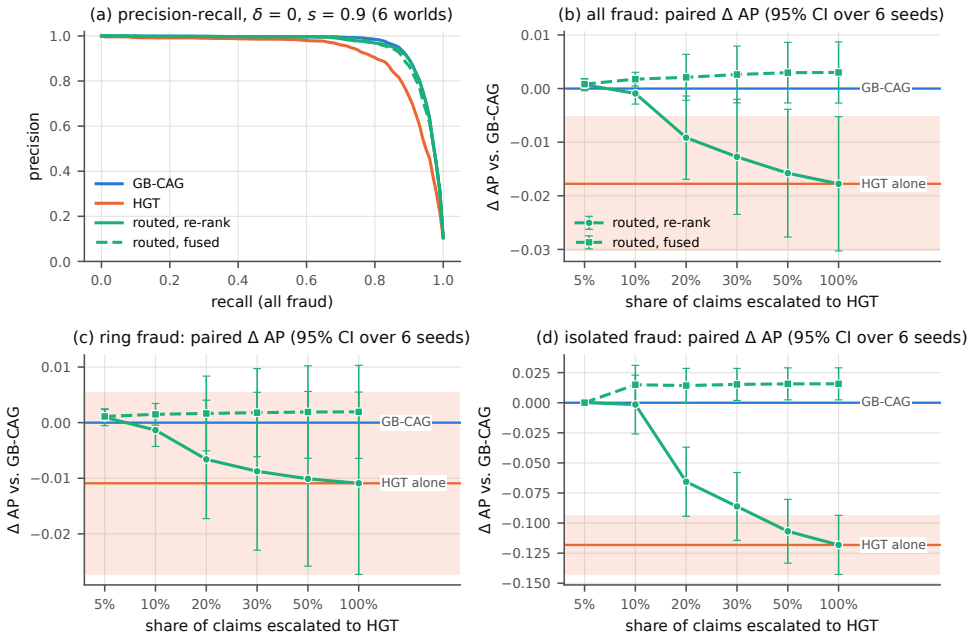


Fig. S1 Two-stage routing from GB-CAG to HGT. (a) Precision–recall curves for all fraud at $\delta = 0$, $s = 0.9$ (precision interpolated at fixed recall and averaged over six worlds; routed systems at the escalation fraction chosen on validation). (b)–(d) Paired difference in test AP from GB-CAG (zero line) against the share of claims escalated, averaged over the 54 worlds, with 95% CIs over the six seeds; orange line and band: HGT alone. The escalation fraction actually used is chosen on validation; the curves over all fractions are shown only to describe the trade-off.

S3 Ring-volume dose-response

Table S3 gives ring AP for every method at four ring volumes (16, 8, 4 and 2 claims per ring per month) at $\delta = 0$, over seeds 1–6, whose legitimate claims, isolated fraud and ring members are identical across volumes. The hyperparameters are those of the main experiment; the re-tuned results at volume 4 are in Table 4 of the article. In a test period a ring patient files on average 9.3, 5.4, 3.7 and 3.0 claims at the four volumes, against 1.6 for other patients (95th and 99th percentiles about 3.8 and 5.2), and a ring doctor 38, 26, 20 and 18 claims against 16. At 2 claims per month a ring files about one claim per member, rings make up 1.1% of the non-isolated claims, and no method is far above chance. Paired comparisons for every cell are in the repository (`results/tables/dose_response_comparisons.csv`).

S4 Confirmatory test of the hybrid result

In seeds 1–6, after re-tuning at volume 4, SAGE+AG had higher ring AP than GB-CAG for sparse rings ($s = 0.1$: 0.129 against 0.078, Holm $p = 0.035$), the only significant advantage over GB-CAG in the low-volume experiments. Because this comparison was singled out after the results had been seen, we tested

Table S3 Ring AP at $\delta = 0$ by ring volume (claims per ring per month; 16 is the default) and structural density s, mean over six worlds (seeds 1–6, which share their legitimate claims across volumes), hyperparameters of the main experiment. Chance level: 0.080 at 16, 0.042 at 8, 0.021 at 4, 0.011 at 2.

	$s = 0.1$				$s = 0.5$				$s = 0.9$			
Method / volume	16	8	4	2	16	8	4	2	16	8	4	2
GB-C	0.084	0.046	0.024	0.014	0.086	0.044	0.023	0.012	0.087	0.044	0.022	0.015
MLP-C	0.088	0.044	0.024	0.013	0.087	0.041	0.024	0.013	0.095	0.046	0.024	0.013
LR-CA	0.923	0.524	0.106	0.025	0.888	0.428	0.087	0.039	0.868	0.364	0.088	0.020
RF-CA	0.875	0.339	0.057	0.019	0.838	0.343	0.064	0.018	0.844	0.331	0.064	0.018
GB-CA	0.925	0.422	0.049	0.020	0.905	0.372	0.052	0.019	0.889	0.311	0.055	0.018
GB-CAG	0.930	0.550	0.066	0.018	0.983	0.879	0.409	0.024	0.990	0.923	0.570	0.019
SAGE	0.938	0.606	0.074	0.016	0.963	0.683	0.108	0.017	0.976	0.755	0.131	0.018
RGCN	0.933	0.546	0.066	0.018	0.953	0.637	0.112	0.018	0.954	0.686	0.142	0.018
HAN	0.908	0.469	0.062	0.014	0.906	0.507	0.065	0.018	0.921	0.520	0.058	0.014
Simple-HGN	0.915	0.561	0.074	0.014	0.960	0.692	0.133	0.015	0.957	0.716	0.151	0.013
HGT	0.910	0.544	0.046	0.016	0.948	0.649	0.079	0.017	0.949	0.689	0.098	0.016
SAGE+AG	0.973	0.659	0.103	0.020	0.985	0.883	0.379	0.032	0.983	0.914	0.566	0.025
HGT+AG	0.964	0.546	0.074	0.019	0.984	0.872	0.324	0.024	0.990	0.916	0.523	0.023
Overlap score	0.254	0.141	0.057	0.018	0.937	0.721	0.194	0.030	0.985	0.866	0.281	0.029

it once more as fixed in an analysis plan committed before the runs started (`paper/hiss_review/analysis_plan_round2.md`): fresh generator seeds 7–18, $\delta = 0$, $s = 0.1$, volume 4, the re-tuned configurations, only GB-CAG and SAGE+AG, and one two-sided paired t-test at $\alpha = 0.05$. SAGE+AG had a mean ring AP of 0.129 against 0.104 for GB-CAG: a difference of +0.024 (95% CI $[0.005, 0.043]$, $p = 0.016$; Wilcoxon $p = 0.009$), positive in nine of twelve seeds, with a chance level of 0.021. By the pre-declared rule the advantage is confirmed; its size is about half the exploratory estimate.

S5 Cost of training and scoring

Median training time per world (Intel Core i7-1068NG7 laptop CPU, two threads per job with several jobs in parallel, so times are indicative) was 3 s for GB-CAG, 39 s for SAGE, 56 s for RGCN, 86 s for HAN, 126 s for HGT and 144 s for Simple-HGN; scoring a test period took at most 0.25 s for any method. A separate single-thread benchmark on an otherwise idle machine (four worlds at $\delta = 0$) measured scoring directly. For one audit period (about 4,900 claims and 11,200 nodes), building the aggregate and graph features took 0.05 s and the heterogeneous graph 0.01 s, and scoring every claim took 0.05 s with GB-CAG and 0.12 s with HGT. Scoring a single claim, as an online system would, took a median of 2.1 ms (99th percentile 3.1 ms, mostly library-call overhead) with GB-CAG on a precomputed feature row, and 10.7 ms (14.7 ms) with HGT on the claim's exact three-hop computation subgraph (about 126 nodes, 1% of the period graph; extraction 2.8 ms, forward pass 7.7 ms), which reproduces the full-graph score to within 1.2×10^{-7}. Routing would therefore cut the online GNN cost in proportion to the share of claims escalated. In batch scoring the escalated claims'

subgraphs overlap: escalating 10% of the claims already touched 55% of the period graph, and scoring that union took 0.08 s against 0.12 s for the whole period. None of these figures includes keeping aggregates, graph features and degree counts up to date as claims arrive, which an online system would need and our code does not implement.

S6 Single-factor ablations of HGT

Table S4 changes one factor of the tuned HGT at a time in the same regime. Depth beyond two layers and the number of attention heads made no significant difference (four layers +0.002; one, two and eight heads −0.037 to +0.013; all Holm $p > 0.5$), so there is no sign of over-smoothing up to four layers on these graphs. Among the relations, removing the patient–claim edges cost most (−0.219, Holm $p = 0.005$); removing the claim–doctor edges cost 0.113 on rings (Holm $p = 0.085$) and 0.147 on isolated fraud ($p < 0.001$), while hospital nodes and the direct patient–doctor relation contributed nothing measurable. The node inputs separate the two tasks cleanly. With claim attributes set to zero, ring AP was unchanged (−0.012) and isolated AP fell to chance (0.036); an HGT that saw only the degree encodings and the graph reached a ring AP of 0.982 ± 0.018, as high as the full model, and detected no isolated fraud. HGT's ring detection is therefore structural—counts of who bills whom, propagated along patient–claim–doctor paths—and its claim attributes serve isolated fraud only. This is the information that the engineered shared-patient features summarise directly. At default volume these counts also carry the ring members' excess claims; Section 6.6 of the article shows that when they do not, the GNNs fall far behind gradient boosting with the engineered features. HGT also varied between worlds (standard deviations up to 0.11), so with six worlds differences smaller than about 0.05 cannot be resolved.

S7 Which inputs the trained models use

To see which inputs the trained models use, without retraining, we permuted one input group at a time across the test period, five times each, and recorded the drop in ring AP: for GB-CAG the C, A and G features and each G feature separately, and for the GNNs the claim attributes, static entity attributes and degree encodings across nodes of each type. At $\delta = 0$, GB-CAG relied on aggregates when rings were sparse (A: −0.80 at $s = 0.1$; G: −0.13) and switched to the graph features as the block became dense (G: −0.81 and −0.88 at $s = 0.5$ and 0.9; A: below 0.06), above all the normalised triadic closure of the claim (−0.47 on its own at $s = 0.9$). SAGE and HGT relied almost entirely on degree encodings (−0.76 to −0.89 at every density) and hardly at all on claim or entity attributes (below 0.05). The two kinds of score are not on a common scale: GB-CAG's measure reliance on engineered summaries of the graph, whereas a GNN's degree encodings are raw counts whose relational meaning is extracted by message passing, and permutation understates groups whose information is duplicated elsewhere. Read together with the ablations, they are consistent with both model families detecting dense rings from the same evidence: how many counterparties a claim's patient and doctor have, and how many of them they share. (Full results, including isolated fraud and test-time rewiring, are in the repository.)

Table S4 HGT single-factor ablations at $\delta = 0$, $s = 0.9$ (six worlds; one factor changed at a time from the tuned configuration: 3 layers, 4 heads, hidden size 32). Ring AP (mean ± s.d.), isolated AP (mean) and paired change from the full model; Holm-adjusted p over every ablation comparison in this table and Table 5 of the article.

Variant	Ring AP	Δ ring AP	Isol. AP	Δ isol. AP
Full model	0.949 ± 0.052	–	0.455	–
Depth (full: 3)				
1 layer	0.788 ± 0.052	−0.161 ($p = 0.020$)	0.239	−0.216 ($p = 0.007$)
2 layers	0.947 ± 0.035	−0.001 ($p = 1.000$)	0.423	−0.032 ($p = 1.000$)
4 layers	0.951 ± 0.043	+0.002 ($p = 1.000$)	0.461	+0.005 ($p = 1.000$)
Heads (full: 4)				
1 attention head	0.920 ± 0.049	−0.028 ($p = 1.000$)	0.384	−0.071 ($p = 0.228$)
2 heads	0.911 ± 0.081	−0.037 ($p = 0.506$)	0.473	+0.018 ($p = 0.813$)
8 heads	0.961 ± 0.023	+0.013 ($p = 1.000$)	0.524	+0.069 ($p = 0.282$)
Relations				
No patient–claim relation	0.730 ± 0.055	−0.219 ($p = 0.005$)	0.311	−0.144 ($p = 0.333$)
No claim–doctor relation	0.836 ± 0.037	−0.113 ($p = 0.085$)	0.308	−0.147 ($p < 0.001$)
No patient–doctor relation	0.902 ± 0.055	−0.047 ($p = 0.413$)	0.423	−0.032 ($p = 1.000$)
No hospital nodes	0.946 ± 0.032	−0.002 ($p = 1.000$)	0.427	−0.028 ($p = 1.000$)
Node inputs				
No degree encodings	0.569 ± 0.110	−0.379 ($p = 0.003$)	0.375	−0.080 ($p = 0.813$)
Claim attributes zeroed	0.937 ± 0.064	−0.012 ($p = 1.000$)	0.036	−0.419 ($p < 0.001$)
Entity attributes zeroed	0.969 ± 0.027	+0.020 ($p = 1.000$)	0.444	−0.011 ($p = 1.000$)
Degree encodings only	0.982 ± 0.018	+0.033 ($p = 1.000$)	0.040	−0.415 ($p < 0.001$)

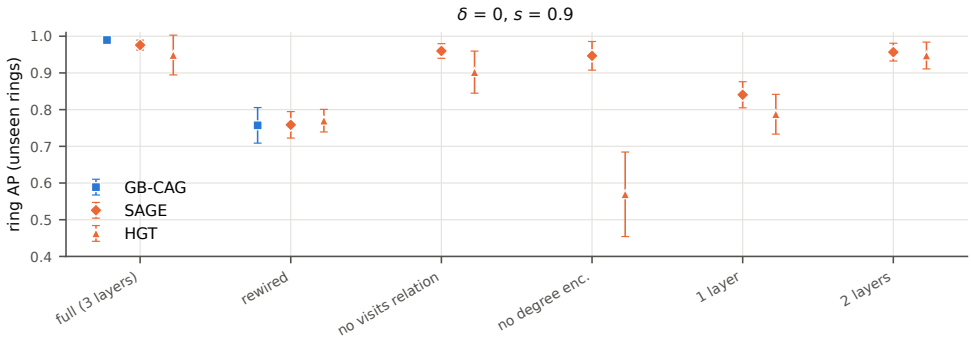


Fig. S2 Mechanism ablations at $\delta = 0$, $s = 0.9$ (mean and 95% t-interval over six worlds). Rewiring preserves every node's degree but destroys the ring block.

S8 Figures of the mechanism ablations and of new versus recurring rings

Figs. S2 and S3 show the results of Table 5 and Table 6 of the article with 95% intervals.

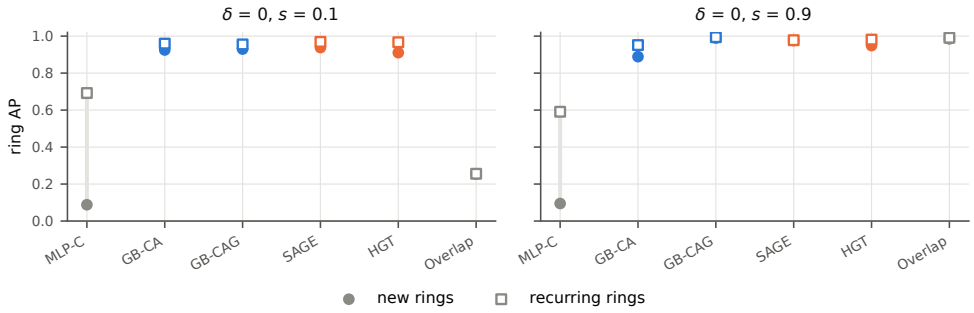


Fig. S3 Ring AP on new versus recurring rings ($\delta = 0$, six worlds).

S9 Repeated cross-validation and per-rotation tuning on the public data

Both analyses were specified in the plan for the second round of experiments before they were run.

Repeated cross-validation. The provider-disjoint five-fold rotation of the article (fold k test, fold $k+1$ validation, three training folds) was repeated with the stratified fold assignment reshuffled four more times (seeds 1–4), giving 25 rotations with the article's split as the first. To save compute, the repetitions covered the tabular models, the MLP and three GNNs (SAGE, HAN, HGT); RGCN, Simple-HGN and the hybrids were run on the article's split only. Hyperparameters are those of the shared selection. Each method is compared with GB-P by the corrected repeated cross-validation t-test (Nadeau and Bengio 2003; Bouckaert and Frank 2004; references 48 and 49 of the article), which inflates the variance of the mean difference by $1/25 + 1/3$ instead of $1/25$, with 24 degrees of freedom, and the eight p-values are Holm-adjusted. Table S5 shows the result. GB-P has the highest mean AP; graph features change it by -0.007 and -0.006, with upper 95% limits of $+0.006$ and $+0.007$; the GNNs are 0.025–0.034 lower. No difference is significant after adjustment (smallest Holm $p = 0.29$, HAN; unadjusted $p = 0.036$). Without the correction, every difference would be significant (all unadjusted $p < 0.01$), which shows how strongly overlapping training sets inflate a plain paired test.

Per-rotation (nested) selection. The shared selection uses a separate 60/20/20 split whose validation providers also appear in the test folds of the rotations. Re-selecting each method's configuration inside every rotation, from the same six candidates and on that rotation's validation fold only, removes this overlap. Table S6 compares nested with shared selection, and every method with GB-P when both are selected inside each rotation. Re-selection changed AP by at most 0.010 for the tabular models and the MLP and by -0.008 (HGT) to $+0.036$ (RGCN) for the GNNs and hybrids, with no change significant; SAGE selected its shared configuration in all five rotations, and SAGE+G inherits it, so their results are identical. We had expected the shared selection to favour the GNNs; it did not do so systematically. With every

Table S5 Public data, five-fold rotation repeated five times (25 rotations; the first repeat is the split reported in the article): AP mean ± s.d., and the difference from GB-P with the corrected repeated cross-validation 95% CI and Holm-adjusted p (variance factor $1/25 + 1/3$, 24 degrees of freedom).

Method	AP	Δ vs. GB-P [95% CI]	p
LR-P	0.674 ± 0.046	−0.019 [−0.044, +0.006]	0.704
RF-P	0.674 ± 0.041	−0.019 [−0.042, +0.005]	0.704
GB-P	0.693 ± 0.045	–	–
GB-PG	0.686 ± 0.045	−0.007 [−0.019, +0.006]	0.704
GB-PGL	0.687 ± 0.046	−0.006 [−0.019, +0.007]	0.704
MLP-P	0.676 ± 0.042	−0.017 [−0.042, +0.009]	0.704
SAGE	0.665 ± 0.058	−0.028 [−0.066, +0.010]	0.704
HAN	0.659 ± 0.046	−0.034 [−0.066, −0.002]	0.289
HGT	0.668 ± 0.050	−0.025 [−0.055, +0.005]	0.704

method selected inside each rotation, GB-P remains the best method, and every GNN and hybrid is 0.023–0.053 below it (smallest Holm $p = 0.051$ for SAGE and SAGE+G).

Table S6 Public data: hyperparameters re-selected inside every rotation (nested) versus selected once on a separate split (shared; Table 7 of the article). AP mean over the five rotations. Nested − shared: change from re-selection; vs. nested GB-P: difference from GB-P when both are selected inside each rotation, with Holm-adjusted p. Intervals are corrected resampled 95% CIs. Same choice: the shared configuration was selected in every rotation, so the results are identical. Hybrids inherit their base model's per-rotation choice.

Method	Nested AP	Nested − shared [95% CI]	vs. nested GB-P [95% CI]	p
LR-P	0.682	+0.003 [−0.008, +0.015]	−0.024 [−0.055, +0.007]	0.607
RF-P	0.680	−0.001 [−0.021, +0.019]	−0.026 [−0.062, +0.010]	0.607
GB-P	0.706	+0.002 [−0.016, +0.019]	–	–
GB-PG	0.693	−0.001 [−0.033, +0.030]	−0.012 [−0.045, +0.020]	0.607
GB-PGL	0.687	−0.010 [−0.040, +0.020]	−0.019 [−0.057, +0.018]	0.607
MLP-P	0.688	+0.003 [−0.008, +0.014]	−0.018 [−0.045, +0.010]	0.607
SAGE	0.664	same choice	−0.042 [−0.062, −0.022]	0.051
RGCN	0.657	+0.036 [−0.014, +0.086]	−0.049 [−0.103, +0.005]	0.527
HAN	0.670	+0.003 [−0.023, +0.030]	−0.036 [−0.089, +0.017]	0.607
Simple-HGN	0.682	+0.012 [−0.032, +0.056]	−0.023 [−0.048, +0.001]	0.527
HGT	0.668	−0.008 [−0.034, +0.019]	−0.038 [−0.065, −0.011]	0.177
SAGE+G	0.653	same choice	−0.053 [−0.078, −0.027]	0.051
HGT+G	0.661	+0.013 [−0.018, +0.043]	−0.045 [−0.096, +0.006]	0.527

S10 Re-analysis of a small fixed benchmark

To show how a small, fixed benchmark can mislead, we re-analysed one: a synthetic dataset of 2,100 claims with five injected rings, produced by an earlier, simpler generator of ours that is included in the repository, evaluated with its own date-sorted

60/20/20 split (Table S7). Such a dataset appears to test ring detection, but this one cannot tell whether relational structure helps, for three reasons. First, its ring claims were drawn from the same inflated distribution as its isolated fraud, so they were anomalous on their own: claim amount alone separates ring from legitimate claims with ROC-AUC 0.94, more than in our most anomalous regime ($\delta = 2$, 0.90), and gradient boosting on three raw claim attributes ranks all 21 test ring claims above every legitimate claim (ring AP 1.000). Second, the same five rings were active for all 24 months, so every test ring claim was billed by a doctor with labelled ring claims in training, which is the memorisation setting of Section 7 of the article. Third, the test split held 21 ring claims (3–6 per ring): one claim moves ring recall by 4.8 points, even a perfect recall has an exact 95% interval of $[0.84, 1.00]$, so a recall gap of a few claims between two models is within noise, and a single dataset says nothing about variation between datasets. None of these properties is peculiar to that dataset. A small, fixed benchmark whose injected rings differ from legitimate claims in their features, recur across the split and contribute a few dozen test claims cannot separate relational learning from claim-level anomaly detection or from recognising known members. Controlling the feature-level signal, testing on new rings and replicating worlds removes all three confounds. Because the generator, features, evaluation unit and tuning differ, numbers from the two set-ups are not directly comparable.

Table S7 A small fixed benchmark (2,100 claims, five rings; an earlier, simpler generator of ours, included in the repository), re-analysed with its own split (date-sorted, 60/20/20), compared with our benchmark. Benchmark ranges span $\delta = 0$ to $\delta = 2$ ($s = 0.5$, six worlds each; ring AP and recall of GB-C pooled over s). Shift and ROC-AUC use raw claim amounts, not adjusted for specialty, for both. Interval: exact 95% binomial.

	Fixed dataset	This benchmark
Claims in total (in the test split / period)	2,100 (420)	20,176 (4,924), mean per world
Ring claims in test	21 (3–6 per ring)	384 (4 rings × 96)
Rings	5, active all 24 months	4 per period, each active 6 months
Test-ring doctors with labelled ring claims in training	100%	0% (by design)
Ring vs. legitimate log-amount shift (s.d.)	1.49	0.11 ($\delta = 0$) to 1.87 ($\delta = 2$)
ROC-AUC of claim amount alone, ring vs. legitimate	0.94	0.53 to 0.90
Ring AP, GB on claim attributes only	1.000	0.086 to 0.811
Ring recall, GB on claim attributes only	21/21 [0.84, 1.00]	0.15 to 0.76
Generated datasets (independent generator seeds)	1 (1)	54 (6)